Quadratic equations

1 The equation $x^2 - 144 = 0$ has a positive integer solution of $x = 12$. Find its other solution.

2 Solve the following equations:

a $x^2 = 2$	b $x^2 = 25$	c $x^2 = 121$	d $x^2 = 294$
e $x^2 - 121 = 0$	f $x^2 - 10 = 15$	g $\frac{x^2}{16} - 2 = 2$	h $\frac{x^2}{25} - 3 = 6$
i $(x+3)^2 = 49$	j $(x-3)^2 = 64$	k $(x-6)^2 = 2$	l $(2-x)^2 = 81$
m $(x-7)^2 = 81$	n $(7-x)^2 = 81$	o $(8x+9)^2 = 256$	p $81x^2 - 16 = 0$

3 Solve the following equations:

a $x(x+7) = 0$	b $x(2x-9) = 0$
c $(10x-9)^2 = 0$	d $(4x-9)^2 = 0$
e $(-3+7x)^2 = 0$	f $(x-4)(x-2) = 0$
g $(x-6)(x+7) = 0$	h $(8x-5)(3x-7) = 0$
i $(3x+8)(5x-7) = 0$	j $(3x-17)(2x-d) = 0$

4 Solve the following equations:

a $4y^2 = 100$	b $25y^2 = 36$
c $-3k^2 = -12$	d $81k^2 + 8 = 24$
e $-25v^2 + 64 = 0$	f $10\left(p^2 - 7\right) = 930$
g $4m(m+5) = 0$	h $\frac{m}{2}(m+5) = 0$

5 Solve the following equation for x, in terms of a and c. Assume a and c are positive.

$$ax^2 - c = 0$$

Applications

6 The equation $4x^2 + kx + 16 = 0$ has one solution: $x = 2$. Find the value of the coefficient k.

7 The equation $ax^2 - 32x - 80 = 0$ has one soulution: $x = 4$. Find the value of the coefficient of a.

8 The Widget and Trinket Emporium has rel[redacted] the forecast of its revenue over then next year. The revenue R (in dollars) at any point in time t (in months) is described by the equation:

$$R = -(t-12)^2 + 4$$

When will the revenue be zero?

9 Neville needs a sheet of paper $x\,\mathrm{cm}$ by $13\,\mathrm{cm}$ for an origami giraffe. The local origami supply store only sells square sheets of paper.
The lower portion of the image below shows the excess area A of paper that will remain after Neville cuts out the $x\,\mathrm{cm}$ by $13\,\mathrm{cm}$ piece. The excess area, in square centimeter, is given by the equation:

$$A = x(x-13)$$

a At what lengths x will the excess area be zero?

b For what value of x will Neville be able to make an origami giraffe with the least amount of excess paper?